

ActInf ArtStream 001.1 ~ "Order and change in art", Sander Van de Cruys,

ArtStream | Sander Van de Cruys, Jacopo Frascaroli | 1:33:06

Hello and welcome. It is April 26th, 2024, and we're in Active Inference Art Stream 001.1, starting a new art-related series, very appropriately, with some great art discussions and presentations. This is going to be Order and Change in Art, with Giacoppo Frascaroli presenting, and also with Sander van der Krooyz and Axel Constant here for discussion. So, thank you all for joining. Looking forward to the presentation and discussion. Thank you, Daniel. Thank you, everyone, for being here. It's great to be able to present to such a knowledgeable audience of Active Inference Aficionados. And the work that we'll be presenting today is a work that Sander van der Krooyz,

Carl Fristone and myself did a few months ago. And it is a work on a topic which I would say is rather neglected or marginal so far in the Active Inference literature. But as I will try to show, it is a topic that is very promising. It has a lot of interesting implications. And indeed, in recent years has been gathering a lot of attention. So let's jump straight into the topic.

And our starting point is the remarkable convergence of interest and results that has been emerging in the last few years between scholars interested in mind and brain from a predictive processing or Active Inference perspective, and scholars interested in studying the arts and aesthetics from a variety of perspectives, philosophy, psychology, and neuroscience, but also increasingly the history of arts and artistic practice as well. So this convergence of interests and results is fostering an ambitious research program that is proving beneficial for both sides. Because on the one hand, predictive processing and active inferences are increasingly seen as frameworks that can illuminate important aspects of our aesthetic experiences. While on the other hand, the arts and aesthetics seen from these predictive processing or active inference perspectives are increasingly seen as potential windows into important aspects of our mental functioning.

So the talk will be exploring these two directions of research. And most of what I will say is taken from these two papers.

One that Sander van de Kruijs, Carl Frister and myself wrote, and that gives the title to this presentation, and the other that I brought together with Sander Helmut Leder and Elvira Brattico.

And the talk papers can be found in this recently published theme issue of Philosophical Transactions of the Royal Society B, which is devoted entirely to these intersections between predictive processing and active inference on the one hand,

and the arts and aesthetics on the other hand. So if you want a sort of overview of the state of the art on the topic, I definitely recommend checking out this theme issue.

The plan for the talk will be pretty straightforward. As I said, we will be exploring these two main directions of current research. The first pertains to the ways in which we can illuminate aspects of aesthetic experience using the active inference.

And in this context, we will be seeing what we mean by aesthetic experience. We'll be introducing the notion

of sense making and epistemic arcs from this active inference perspective.

And we will be talking about how these notions can clarify certain aspects of our engagement with the arts and our aesthetic experiences.

In the second part of the talk, we will be talking about ways in which, on the other hand, our mental functioning can be illuminated by the arts and aesthetics considered from this active inference or predictive processing perspective.

And we will be examining a sort of analogy between what artists do and what cognitive scientists do when they study the mind.

And we will be clarifying how an examination of the arts and aesthetics can throw light on, among other things, the dynamics of inference, the dynamics of affect and certain aspects of well-being of psychopathology.

And in the end, I will be presenting some conclusions in some directions for future research as well.

So let's start by saying something on how we can illuminate aesthetic experience from an active inference perspective.

Now, we should probably start by clarifying what's what is our explanatory target here?

That is aesthetic experiences. Aesthetic experiences have been characterized by the philosophical and the psychological literature as experiences traditionally associated with the notion of beauty, the perception of beauty in some sensory array.

And these are experiences that we undergo with varying degrees of intensity in the most varied circumstances of our life.

It could be that we are looking at the wonderful vista, that we are examining and finding pleasure in the arrangement of furnitures in a living room.

But most characteristically, aesthetic experiences are experiences that we undergo while engaging with art, while listening to great pieces of music,

while confronted with great work of visual art, while reading great novels and so on.

Now, there is a set of features that are recurrently attributed to aesthetic experiences.

And that's the feature. These are the feature that we we are trying to explain from this active inference perspective.

First of all, aesthetic experiences are considered to be pleasurable. These are experiences that are accompanied by the pleasant, effective response normally.

This doesn't mean that they are not uncomfortable or riddled with difficulty. But in the end, overall, they tend to be pleasurable.

They are also self-sustaining. And with this, by this, I mean that they are experiences in which we seem to engage without any further purpose,

and in which we seem to engage in a continuous manner, finding in this particular sensory array that we are examining,

and to be able to engage in a different way, and to be able to engage in a continuous manner,

and to be able to engage in a continuous manner, and to be able to engage in a continuous manner.

And so they sustain our serious attention throughout, and we seem to care about them without any further utilitarian purpose in mind.

Extending experiences are often said to be transformative, in the sense that when the painting is behind us, the music is over,

when we put down the book or leave the theatre, we feel that we are not the same as before, that we have changed and usually changed for the better.

This is a particularly strong trait associated with aesthetic experience that many refer to, and it can happen, of course, with various degrees of intensity.

It could be that you just felt like your worldview has changed slightly, or it could lead to massive revelations about anything, basically.

Aesthetic experiences are also said to be subjective, in the sense that what is aesthetically appealing or pleasurable to me,

may not be aesthetically appealing or pleasurable to you.

They are also said to be transient and evolving, in the sense that they are characterized by an evolving affective and cognitive profile,

that culminates normally with a pleasurable response, but it's not that pleasurable response or judgment alone.

They are also evolving in the sense that what I tend to prefer could change for me in time.

It's not just that what you prefer and what I prefer are different, but also what I prefer in this moment in time can be different from what I prefer in another moment in time.

And also aesthetic experiences tend to be seen as related in some way with a feeling of freedom from existing constraints in our perceptual, cognitive or motor processes,

and a feeling of enhanced perception of our possibilities for action.

We will see what we will say more about that, but these are more or less the features that we are, that we choose to concentrate on.

And there are more characteristic of aesthetic experiences.

Now, the way in which we go about trying to explain these characteristics is by introducing these two notions and trying the notion of sense making and epistemic.

epistemic arcs and trying to characterize them from an active inference perspectives.

So to understand them, let's start with an example.

You've probably seen pictures like these before.

They are quite common in the psychological literature, but perhaps you haven't seen this particular picture before.

If you haven't, it will probably look like a random array of white and black patches at the beginning.

But if you concentrate enough on it, after a while, you may start to notice that there is an animal in the picture, more specifically a cow.

If you can see the cow, here are some cues in the picture.

You just see the face of the cow.

These are the nostrils of the cow.

These are the eyes.

These are the ears of the cow.

Now, if you see the cow now, you did something pretty remarkable already in the sense that you have

arranged what seemed to be like a random array of sensory stimulation into something meaningful.

And that's what we mean by sense making.

And we can like sort of break down this process from an active inference perspective by thinking about a sort of epistemic arc involved in the discovery of structure in our sensory stimulation.

We start with uncertainty.

Uncertainty about the potential causes of our sensory stimulations.

These uncertainty in some cases, and this case, for example, triggers an expectation of the possibility to reduce such uncertainty.

And that's what curiosity looks like from an active inference perspective, according to an increasing literature.

There is curiosity, the feeling that we can find structure in our sensorium, that if we look for enough time, we will be able to understand something is characterized from an active inference perspective as expected uncertainty reduction or expected information gain.

Which means the extent to which we predicted we will be able to reduce uncertainty about the causes of our sensory stimulations.

If we are curious, that is, in the cases in which uncertainty is such that we seem that the stimulus sort of yields a promise of making sense if we concentrate on it enough.

We will probably try to put in place epistemic actions, that is, actions that we perform in order to reduce uncertainty.

And this could go from saccades, as we examine the picture, up to internet searches and every source of actions aimed at reducing the ambiguity or the uncertainty of our sensory stimulations.

And if we are lucky enough, these epistemic actions lead us to uncertainty reduction.

And in the end, we experience pleasure.

You may have noticed, you will have noticed that when you found the cow in the picture, as with many of these Mooney images, you will probably felt a positive affective reactions.

The psychologists normally call an aha moment or an aha experience that has both cognitive and affective components and that accompanies the increase, the sudden clarity about the causes of your sensory stimulation.

So overall, this experience generates a sort of arc, as we said, that goes from uncertainty to pleasure.

And it is an arc. It is important to stress that this is an idealized experience because things could go wrong in many ways.

For example, uncertainty in your sensory stimulation could be so high that you are not quite sure that you can reduce it.

And so you don't experience curiosity in the first place because you are not confident in the possibilities of reducing uncertainty in your stimulations.

But it could also be the case that your epistemic actions do not lead to uncertainty reduction, but to an increase in uncertainty instead.

After all, it's a changeable world.

And even if we put in place actions to reduce uncertainty, it could be it could well be that uncertainty stays the same or increase even.

And things could also go wrong because uncertainty in some cases is not reduced to an appreciable extent

or is not reduced fast enough for us to experience this pleasurable, effective response.

So as I said, these epistemic arc is an idealized experience, a way of capturing successful sense making, we will say.

Sense making that happens in an optimal way, but things could go wrong in many, in many possible ways.

Now, if we think about why we experience pleasure while guessing the structure of our sensorium, while settling on a specific hypothesis about the hidden causes of our sensory data.

Well, that has a pretty straightforward explanation from an active inference perspective, because in this perspective, as we know,

reducing uncertainty is pleasurable because it means maximizing model evidence that is maximizing the evidence for your own existence, something which how and others have called self evidencing.

But it's important to stress and we stress in the paper that self evidencing is not to be seeing a sort of passive process of confirming already preexisting belief,

but it's more like an active process that is a process of self transformation as well, because in finding the structure of our sensorium in defining it in determining it,

we are also determining the structure of yourself as a model of that sensorium.

And so finding structure is always restructuring yourself in this way, which is a very active process.

And that will be relevant in the, in the, in the, in the rest of our discussion.

So reducing uncertainty is pleasurable.

And these grounds, several theories of affect and wellbeing from a predictive processing or active inference perspective.

I put some of the papers, relevant papers there.

They all more stress that increasing prediction error or are normally affectively negatively balanced, whereas decreasing prediction error are normally seen as positive.

And these links well with certain assumptions and certain empirical results in the psychological literature and in literature in developmental robotic, for example,

it is known for example, that infants tend to concentrate on stimulations that are neither too unpredictable, nor too predictable,

because these are the stimulation that maximize their learning and their, their refining of their model of the world.

This is known as the Goldilocks effect.

And these sort of environment is spaces of the, of these regions of the input space that afford optimal uncertainty minimization.

I said, sometimes called progress niches, and we'll see the relevance for that or art in a minute.

So overall, there is a view that affects several aspects of our effective responses are related to the dynamics of uncertainty minimizations,

and that we experienced positive effects in relation to our successful attempts at uncertainty minimization.

And that these things makes us concentrate towards regions of the input space that afford such an optimal reduction in prediction error minimization.

Now, let's try to apply all this to the case of art.

Our contention in the paper that Sander has been developing for some years now,

is that art provides us with good epistemic arts.

That is art provides us with these idealized experiences of uncertainty minimization, whereby we start with something which is ambiguous or not responding immediately responding to our predictions.

Then we, these things ask us to perform epistemic actions that are in the end successful and that leads to a pleasurable response.

So if we, for example, take this painting by Monet, we see this process going on at sort of low level, a perceptual level, which is that the ships and the water here appear so vividly because we probably put in place the same thing, the same process that we put in place in the case of the cow.

That is, we have some ambiguous simulation, something which not conform to the way in which normally water or ships look to us.

But if we guess hard enough, we are able to see water and ships, and then we get these aha moments, this pleasurable, effective response.

So what Monet is doing here at a low perceptual level is building such an idealized form of epistemic arc.

But of course, that's an important point, is that these epistemic arcs can go on at different levels of abstractions.

And here we can exploit the sort of hierarchical structure that active inference and predictive processing insist on.

For example, in certain paintings like this Jan van Eyck painting here, the matter is not guessing what figures are in the painting because these are represented realistically, but the point is to guess the meaning of the various attitudes, the various characters in the picture, or the various elements that the painter represents, the specific position in the painting as well.

So these are higher level guesses, which if successful lead to the same circle from uncertainty to pleasure that we saw on a perceptual level in the Monet painting.

And of course, these guesses are not just happening at different levels, but they are also different kinds.

For example, in the case, for example, of expressionist art, the matter is not just to guess the meaning of the painting or the objects represented in it, but to a larger extent to guess the intentions, the state of minds that the painter was experiencing, or had while depicting himself in this case or any subjects.

And this is a form of guessing of other mental states that is, of course, very important in the predictive processing literature, and which expressionist art in particular leverage a lot.

But other guesses can have to do, for example, with the sort of movements that are implied or were put in place by the painter in order to produce the painting.

We have an example of the famous cuts by Lucio Fontana here, but you can also think about the thick brush

strokes by Van Gogh, for example.

These allow these sort of artworks allow the viewer, the viewers to make guesses about the actions that went into the production of the artwork itself.

So as you see, the predictive processing active inference apparatus, if leveraged to the full extent of its possibility,

can account for the pleasure that we get from different artworks in different styles in different moments in time as well.

Now, it's also important to understand that these guesses at various levels inform the guesses at the others level.

For example, if you have understood more or less what figures are in the painting and you're making high level hypotheses about their meaning,

and you make a high level hypothesis about something in particular, you might be pushed towards re-examining certain portions of the painting and noticing something on a perceptual level that you didn't notice before.

So often in art, solving ambiguities or uncertainties at a lower level opens up problems on a higher level, and solving problems on a higher level reopens problems on a perceptual level.

And these in successful art goes on and on and on and on and prolongs our engagement with the artwork itself.

Importantly also, artworks, great artworks in particular, are such that they keep us guessing.

They have a good supply of structural indeterminacy, as a Uri Lothman puts it.

That is, they are ambiguous in many ways and such that if you solve the ambiguities the first time that you see it and you experience certain epistemic acts,

if you revisit it, you might experience others epistemic art that you hadn't considered before.

And you can even experience more pleasure as a result.

But indeed, great artworks are seen as those sorts of inexhaustible stimulations that keep providing us with important insights.

And that might be a way of characterizing how they do so.

In sum, if we take into account all these ways in which art keep us guessing,

we will see that artworks are conceivable, are interpretable as hyper-structurable sensory streams,

progress niches in their own right that allow us to keep improving our models of the world in very refined way.

And perhaps then the attraction that we have towards these stimulations is just part of a broader attraction that we talked before

for regions of our input space that allows our optimal cognitive functioning.

Now, in this picture, it is important to stress that aesthetic experiences are subjective in the sense that whether a specific sensory array will afford a pleasurable experience for a specific agent will depend on the generative model that it commands.

What is unpredictable or ambiguous or uncertain for me can be perfectly predictable for you.

And so if that's the case, it doesn't prompt the same epistemic arc that it prompts for me.

And we see these constantly in the development of individual and collective taste that often demands that we search for more ambiguous,

more unpredictable stimulations as we learn more about a specific art form in order for us to recover the sense of wonder and insight that certain particular simulations have.

with the right kind of ambiguity gives us.

In this picture, aesthetic experiences are also transient and evolving in the sense that they depend on the specific dynamics of uncertainty reduction and epistemic arcs involved.

And so, for example, for me, this Molle painting, if seen many, many times, won't probably cause the same feeling of insight and pleasure that it caused the first time when I guessed the presence of the ships and the water in the painting,

much like seeing the cow for the second time, the cow that you saw before, doesn't prompt the same aha experience to the same extent,

because now you are more likely to directly interpret it as a cow in an automatic way.

And so you see that aesthetic experiences are even with the same stimulus can be very transient and evolving.

And in a sense, we are always trying with aesthetic stimulations to recover the sort of ingenuity that we had while seeing the picture or the stimulation for the first time.

And art is often characterized as providing us occasions to see things as for the first time.

And that could be a way in which it does so.

Now, the last thing I will say about aesthetic experience from this active inference perspective is that,

as we saw, aesthetic experiences are often characterized as having something to do with freedom of enhanced or enhanced sense of agency.

For example, Kant talks about the free play of cognitive faculties that will be involved in the perception of a beautiful object.

Schiller even goes as far as saying that it is only through beauty that man makes his way to freedom.

So there is an ancient philosophical view that links aesthetic appreciation, aesthetic experiences to freedom from existing constraints and enhanced perception of our possibilities for action.

How do we explain that from an active inference perspective?

Well, there are several ways of accounting for that aspect of our aesthetic experience.

One is that if the picture that we gave so far is right,

aesthetic experience are occasion of enhanced self structuring in the sense that they are moment where we determine the structure of our sensorium at a faster rate than usual.

And in doing so, we determine our own structure in a faster rate than usual.

In a sense, we keep building ourselves, we keep changing, we free ourselves from existing ways of perceiving things,

and we turn our minds into moving targets while we engage with these stimulations.

So these probably capture something about the notion of autonomy that Kant had in mind also when speaking about art.

And that is intriguingly finding new applications in cognitive science as well.

And activists, for example, insist on this notion of autonomy as well, which is giving yourself your own rule.

And perhaps in aesthetic experiences, we are doing that to an enhanced degree because we are determining the structure of the sensorium and the structure of ourselves as models of that sensorium in a faster rate, at a

faster rate than usual.

But another way of thinking about how aesthetic experiences can boost our sense of freedom and agency is that these experiences are normally conducted or encountered in the absence of pressing needs.

That is, we have no pressing needs that push us to organize our sensorium in precise ways.

And we are free to explore many hypotheses about the underlying structure of our stimulations.

And this is probably what people in aesthetics talk about when they talk about the aesthetic disinterestedness.

That is, that specific state of mind that one has to have when approaching something aesthetically, which is a state of mind or a mind of disconnect with practical interest or interest of other kind.

And this frees up space to explore other possibilities of perceptual and cognitive organization.

Another thing that one can take from the active inference framework in order to explain this relationship between aesthetic experience and freedom is that perhaps as we speculate in the paper,

when we experience a sudden reduction in prediction error or uncertainty,

in uncertainty, these in a certain sense boost our confidence in the ability to reduce uncertainty in the future.

There are interesting studies by you and colleagues, for example,

or Barbieri and colleagues that show in this case, for example,

that even an aha experience of the kind that you experience with a cow can encourage risky decision making in a subsequent task.

And this study by Barbieri and colleagues shows that the stimuli that we tend to like are also accompanied or foster more curiosity.

So in a sense, one can see successful aesthetic experiences as something that allows us to broaden that epistemic heart that we examined before.

And because we have more confidence that we will be able if we engage with epistemic actions for subsequent for the necessary time to reduce uncertainty.

And we all face experiences in which these epistemic arcs are very short.

In the paper, we do we make the example of Internet memes, which are for this very short epistemic arcs of uncertainty and discovery.

But art is normally seen as boosting our confidence in our ability to endure and uncertainty, something which is sometimes referred as negative capability.

And so that's perhaps enough, an interesting effect that art could have if examined from this active inference perspective.

Now summing up this part of the talk, we have seen that the active inference or predictive processing apparatus helps us explain why aesthetic experiences are pleasurable,

namely because they are instances of successful self evidencing,

because they are made possible by sensory streams that fuel our curiosity and keep us guessing.

They are transformative because they are instances of enhanced self structuring.

They are subjective because they depend on the agent specific generative model.

They are transient and evolving because they depend on the specific dynamics of uncertainty reduction.

And they are linked to freedom and enhanced agency because they are instances of enhanced self structuring again,

conducting in the absence of pressing needs.

And that boost our confidence that as uncertainty minimizes.

Now for the second part of the talk, which is slightly shorter,
we will examine ways in which our mental functioning can be illuminated by the arts and aesthetics,
if we consider them from this predictive processing perspective.

Now, it's important to see that in this framework,
artists are seen as experts in cognition and in inference in particular,
in the sense that they know what cues ought to be presenting to our sensory organs for us to formulate
certain evolving hypotheses about what we are sensing.

A painter, for example, knows what sensory array to produce in order for you to guess that there is a certain
object in the painting,

but not just that, in order for you to guess it with a certain profile of uncertainty minimization.

And artists have sort of implicit, largely implicit knowledge of these dynamics of uncertainty minimization.

For example, they can feel the tensions, the visual tensions of an imbalanced picture.

They can tell what notes or what word we expect in a sentence or in a melody,
and they can craft cinematic flows with the right kind of abruptness or invisibility in the editing, for example.
So every artist has in this framework sort of a large body of implicit knowledge about the dynamics of
inference.

And indeed, art is probably a place where these dynamics of inference are in their cognitive and affective
correlates become particularly visible in the sense that all the what artists are doing is to manipulate these
dynamics in order to produce on us cognitive and affective reactions.

So therefore, the thesis is that artists and aestheticians to the extent that they have access to this large body
of implicit knowledge might therefore contribute to our understanding of modern things,
the dynamics of inference, the dynamics of affect and certain aspects of well-being and psychopathology.
So let's examine these three things in turn.

Well, as we said, probably artists possess this implicit understanding of our inferential dynamics.

For example, if Mozart here crafts a melody, he knows whether a certain note is to be expected given the
preceding context or whether to introduce another note,
which is not expected given the predicted context in order to generate certain effects.

In fact, some people characterize music theory as the most developed psychology, folk psychology currently
extant, exactly to capture this fact that in within artistic practice,
there is embedded a large body of knowledge about the dynamics of our inferences.

And the same goes for a poet, for example, this is not a particularly poetic example, but for a poet that
chooses carefully the words of his poem in order to generate certain profiles of uncertainty minimization.

So one can say that artists are astute manipulations of the inference we make and the timing with which we
make them.

And this knowledge can indeed be exploited to link the personal and the sub-personal level when we
examine experience, for example.

And this is an interesting point.

Many of these things interestingly are starting to be examined from a predictive processing perspective and again art could be a way in which we bring the expertise of artists and aestheticians about these states and the way they are produced to bear on research in cognitive science that examine the same thing and this could also inform debates in philosophy and cognitive science about things like affective engineering, affective scaffolding and extended affectivity. There is ways of conceptualizing the many ways in which we modify our affective states by modifying things in the environment and we manage our affective states by managing the sort of stimulation that we encounter. Art seems to be a particularly interesting case of affective scaffolding or extended affectivity and it is interesting to see that indeed art is starting to be examined from that perspective in some recent work as well. And finally as for well-being and psychopathology, well we saw that aesthetic experiences, if we are right, have a lot to do with optimal cognitive functioning in the sense that they afford those optimal experiences that we characterize as epistemic arts. They are about optimal information seeking, about optimal development of our mental models. So to the extent that artists and aestheticians have access to these dynamics, perhaps we can gather for them indication on how to design experiences that favor involvement, motivation, learning and discovery. And this is indeed an increasing stream of research that goes sometimes under the name of art therapy. So to the extent that artists and the extent that artists have been able to do with art therapy,

and there are many studies examining the effects of the exposure to artistic stimuli on several cognitive faculties and abilities that seem to go in that direction, in the direction of suggesting that artists have some sort of knowledge on how to arrange sensory streams that boost these things, involvement, motivation, learning and discovery.

And finally, perhaps, if aesthetic experiences have a lot to do with optimal cognitive functioning, they can perhaps shed light on forms of pathological cognitive functioning or epistemic behavior.

Because indeed, one thing that active inference allows us to do is to conceptualize many pathologies of our collective information seeking, like echo chambers, conspiracy theories, and so on, and many psychopathologies like schizophrenia, delusion, autism, anxiety, and mood disorders like anxiety and depression, is having to do something with suboptimal ways of engaging with environmental uncertainty.

So if art can tell us something about optimal cognitive functioning, perhaps it can tell us something about deviations from this optimal cognitive function, which is important to stress, are often put in place in order to recover that viability in the end.

And on the other hand, studies about these pathological conditions could also inform the production, the study of the production of art and aesthetic experiences.

So let's draw some conclusions now and some direction for future research by just reverse rehearsing what we said so far.

So we have seen that there is a remarkable convergence of interest and results between scholars working within the predictive processing or active inference framework and scholars studying the arts and aesthetics.

This convergence is proving beneficial for both sides because on the one hand, predictive processing and active inference are illuminating aspects of our aesthetic experiences, things like why they are pleasurable, why they are self-sustaining, transformative, subjective, transient, and linked to freedom and enhanced agency.

Apologies for the misspelling there.

And on the other hand, art and aesthetics are beginning to be seen as windows into mental functioning and revealing something about the dynamics of inference, the dynamics of affect, and wellbeing and psychopathology.

Now to point out directions for future research, in aesthetics in particular, which is a big field that has contributors from philosophy, psychology, neuroscience and so on.

The next thing to study probably from this predictive processing or active inference perspective or the next thing that are needed are more empirical testing of the predictive processing, hypothesis which are sometimes very difficult to test.

In the introduction to the theme issue, we examine some of these difficulties which have to do with mainly with the way of gathering measures of uncertainty and uncertainty minimization, for example.

So more empirical testing is needed, but also more insights into the neural underpinnings of these dynamics dynamics which are quite often linked to the limbic systems and the relationship between the way in which we

conceptualize dopamine

and other hormones from a predictive processing perspective.

But the neural underpinnings of these epistemic us, these aha moments are still all to be clarified.

Also, it is important to understand how these predictive processing perspective that we are advancing connects with links to other frameworks in empirical aesthetics,

which is a big field with a long history.

And this is particularly important if you want to discern the specific predictions that have to do with the predictive processing framework

from predictions that are made in other frameworks as well.

And an important also expansion of the story would be its extension to not just exteroceptive stimulation,

which have been the subject of our attention in the presentation and in our papers mostly,

that is stimulation that we get from the environment, but also proprioceptive and interoceptive

stimulations that have also their own inferential dynamics.

And they could open possibilities, for example, to conceptualize also things like sports,

performative arts, the playing of instruments, for example, or skillful motor behavior

as a way of as a form of aesthetic experience itself.

And another relevant expansion of the story, which hasn't been covered so far a lot,

is taking the artist's perspective, because we talked a lot about aesthetic experience,

that is, experience of subjects exposed to aesthetic stimulations.

And we haven't talked much about the way in which these stimulations are produced.

That is, the creative process that is put in place by artists in order to generate these carefully curated sensory streams.

And there is a very interesting beginning of that exploration in the paper by Axel Constant and colleagues

in the theme issue, which I recommend, I refer you for more on creativity and active inference.

More broadly, the expansions of the view that I are presenting for cognitive science

have to do, as we said already, with studying the dynamics of inference after the well-being of psychopathology,

which are all directions that are very much still to be explored.

But also, there are many other phenomena that are normally examined under an aesthetic perspective,

but they have also been given accounts from a predictive processing perspective,

things like learning, attention, motivation, and conscious experience,

that are very much still to be examined from this perspective,

and the need, and they would probably profit from the encounter between aesthetics and art on the one hand,

and active inference on the other.

Okay, I conclude by just reminding you again of the theme issue on these topics,

and thanking once more my collaborators, first of all, Sander van der Koers,

which is also here, and will be able to stay for the Q&A and answer your concerns as well.

Carl Fristen, Helmut Leder, and Elvira Bratiko.

Thank you very much for your attention. I look forward to an interesting discussion.

Thank you. Thanks for the presentation. Awesome.

Thank you. All right. Well, lots of places to jump in. Perhaps Sander first, and then Axel, if you would like to make any initial remarks, or just give any context, or what you are left with from the presentation.

Well, it was a great overview of the work we've done together, and

I have a little to add immediately. But yeah, I'm interested in, like, how can we empirically test this, and how can we quantify uncertainty in a way that we can link it to people's experiences.

And in music, this is happening a bit, like people are using models that output measures of uncertainty while processing progressions of chords, for example. This is the work of Cheng.

And there it's found that these models, the surprise measures and uncertainty measures that are derived from these models, they

are using the same, like, how can we predict the emotions that individuals are experienced when listening to this music. So that's an interesting,

interesting, an interesting step. And next step, I guess, would be to look at it at a more,

more personalized way in which we, for example, have jazz lovers, and we train this,

this computational model on jazz music, and we see whether the surprise and the uncertainty that that kind of model outputs that that it relinks better to the emotions of the jazz lovers,

than to general listeners. So then it's really tuning the uncertainty to the individual or the group in

this in this case. So that's, for me, a very promising and interesting route to take next. But in

general, I think I'm happy with with the story that Jacopo told. So I want to hear your reactions.

Great. Axel? Yeah, well, thanks Jacopo for the talk. It was really interesting.

So yeah, I guess I have two large sets of questions, one of which you already sort of

highlighted, which is about creativity, right? Relation between the creativity as

the creativity as that which as an attribute of the artist's

product or process. And that's how you sort of operationalize it. In psychology, sometimes you will

have the creativity of the thing that you sort of produce that is the creative product or a creative

process. And that is my my experience as someone who produced that thing that is creative, you know,

that can also be creative in a sense, right? And the first or the set of questions that have to do

with creativity for me, it's especially this interesting observation that the creativity of the

process doesn't seem to be predictive of the creativity of the product, right? Because product

creativity, you have two criteria, the aptness and the novelty, right? And the aptness has to do with,

let's say I want to create a stand for a laptop, it can be a creative laptop stand. But for that stand to

be creative as a product, quote unquote, it also has to work, also has to hold my laptop, right? So that's

the aptness criteria has to be useful, and has to work for what it's designed is designed to do,

right? And then you have a novelty, which is a statistical criterion, which is from the point of

view of all the kind of laptop stands that have been produced, which of those are considered most

original, if you will, from the point of view of the people who would assess all the population of

laptop stands, right? And again, that doesn't seem to track with the creativity of the process of me producing laptop stands, right? So I can think or feel like I'm producing a pretty boring laptop stand, and it turns out to be a very creative product, creative laptop stands, right? So I would be interested in hearing you on this sort of, you know, not a contradiction, but something that is slightly unexpected, right? So that's the first set of questions concerning creativity, it revolves around this finding interesting finding. And then the other set of questions has to do with, I think this is a more general problem in the predictive processing literature, and aesthetic aesthetic experience. We often say that for you, it's the concept of the aesthetic arc, right? And hearing you, it sounds as though traveling on the arc, if you will, so reducing my uncertainty would in and of itself be pleasurable, right? And I'm not very clear on why that would be the case. I would like that to be the case, because it sounds very intuitive, right? When I reduce my uncertainty, it's like it's a, you know, it's good. I like that it works well, right? But really, for that to be the case, there would be, we would need kind of a story about how the computational neuro, computational physiology, according to active inference, right, maps onto the actual physiology, neurophysiology of pleasure, right? So I'm not sure we have that story. Well, maybe we do, but I haven't seen it, right? Or the alternative would be something like, once I'm at the end of that sort of travel on that arc, right, I arrive at an outcome, which will be pleasurable or rewarding. And that works for me, right? But the arc in and of itself, because it's sort of, it's a process of minimizing uncertainty, it would be therefore pleasurable. I would need more explanation of why that would be the case, right? So I guess, yeah, those are the two sets of questions.

Thank you. Excellent questions. I'll probably start if you don't mind, Alessandra, and then you'll probably add more, especially perhaps about the second question as well. But in any case, yes. So creativity, process creativity versus product creativity.

I think the discrepancy that you point out could be characterized as thinking, again, about the fact that people have different general command different generative models and have, for example, this could lead to different views about what's a great way of arranging a sensory array in order to produce certain results. An artist could feel that especially if it has practiced similar procedures all over very frequently, that the solution at which he or she arrives at is boring or uninteresting.

And the audience may feel very different for the fact that the generative model is different. They don't expect a particular solution.

Perhaps even it's often the case that this solution has been arrived at serendipitously, that the designer hasn't accounted for certain uses or certain way of reading the stimulation that people find in the simulation themselves because they command different generative models.

And so perhaps there are ways of accounting for this discrepancy between product creativity and process creativity.

But it's very interesting. I've always found interesting the way in which product creativity is described, as you said, as having to do with both aptness and novelty,

because it seems that these two things captured two aspects of the story that we have been telling.

novelty of course, captured the uncertainty, unpredictability of a certain sensory array

and the way in which you should interact with it.

Whereas aptness captured the end of the parable,

when you get the insight and you get how all the pieces should fit together.

And you need to have both in order to have an aesthetic experience in our account and in order to be creative.

So perhaps there is the nice there is the nice link here between the aesthetic experience of the viewer and the creative process that generated that experience.

Right. So it would be like I had this laptop stand that I look at it and I'm like, what's that?

And I start trying to use that thing and then I figure out that it is a laptop stand and that whole process.

Right.

You know, once I figured out that it is a laptop stand and it works well as a laptop stand, right, the aptness criterion is met.

But I had to get to that point.

I had to go through a process of uncertainty minimization because it was such an original and novel.

Exactly.

Because you notice aptness more in a sense if it comes into a context in which you need to discover it.

Otherwise, you are just acting automatically on an object that you already know how to use.

You can perceive the creativity of the object.

Even a mug is creative in a sense because it affords a certain way of using it.

But you need to recover in a sense that creativity that past generations of makers have put into that object, which is not visible to you anymore in a sense.

So again, you have a sort of relationship between creative processes and which looks create look or do not look creative

according to how habituated we are to them and the aesthetic experience as we conceptualize it,

which has to do with stimuli that are novel enough for us to notice them and but not so novel that we are not able to use them anymore.

Yeah, I would just add to that because that's very important to see like the aesthetic experience also as a creative act in a sense

that this blurring of the line between the creator and the perceiver is important in this view that every aesthetic experience is in a sense because of this discovery aspect and a creative act as well.

And so the ethnos is very much dependent on the models of the of the viewer or the models of the creator.

But I mean that there is discrepancy outside and inside.

That's like a logical thing to me.

And of course, as much as the models overlap between these the audience and the artist and then there will be agreed upon creativity strength as it were of the the object.

But but it might come across as very creative for one person, but not not at all for another person.

So, yeah. Yeah. And it.

You know, you often hear stuff like especially for things like contemporary art, right, where people will say, I don't see the point.

I don't see why that's interesting. Right.

Whereas for another person who might be somewhat of an expert in contemporary art, that person would see how interesting a novel as a product that piece of work is.

Right. So it sounds like even though you can explain why a creative process does not necessarily lead to a creative product.

Right. But there must be some kind of, you know, there has to be some kind of, you know, you know, tracking or correlation between the two models, the model of the perceiver and the model of the artist that has produced the product or a piece of work, because otherwise you fall into those situations where people they just don't understand what's going on. Right.

And then there's no aesthetic pleasure or aesthetic experience. Right.

So then the question is, you know, is there a sweet spot of kind of mismatch between because that would that would be exactly what accounts for the possibility of an aesthetic art. Right.

There must be some kind of mismatch between the model, the artist that produced the product and the model, the perceiver. Right.

That that, you know, perceives the product. And because if there is no mismatch, it would just be like a clear communication, like a stop sign. Right.

The model of the government that produced the stop sign matches perfectly your model of what a stop sign is.

And so there isn't really a sort of an aesthetic arc to go through to understand the meaning of the stop sign. Right.

So the question becomes, what is the sweet spot basically in terms of mismatch between the model of the artist and the model of the perceiver?

Right. And then maybe you can look at art history and what typically is the sort of educational background of great artists?

What kind of typical experience they had in their life and what kind of model that could have led to and how mismatched that type of model?

Right. Is with the model of the population at a certain time in history. Right.

So, yeah. Yeah. Yeah. And it's I mean that it's also like the culturally dominant model that is important there.

And you see that artists will play around and will hit and miss and these kind of things.

I mean, they will experiment a lot and be early on their own times.

Like they might propose like some forms of uncertainty that culture cannot deal with at that time.

And a lot of these those things we just don't know about, of course, because they're not filtered by history.

It's only afterwards that we we realize like these dynamics were play were playing and he found the sweet spot.

But but yeah, it's not in advance, I guess, or very hard in advance to.

To predict.

Yeah, I would add that there is a whole field like in aesthetic theory or in the theory of art, which is called like the aesthetics of reception, which whose role is exactly to study the sort of system of expectations that a

certain audience has at a certain moment in time.

And the way in which the new artistic product interacts with these expectations by violating them to a larger or lesser extent.

And so there is already a tradition of studies that understood that if you violate things too much, the reception is not going to be good.

If you violate them too little, it's not going to be interesting.

So this sort of sweet spot talk that you that you adopted is very much used in that literature as well.

And of course, this active inference story has the merits of connecting those early intuitions with these neurocomputational language and and all the literature on all the other cognitive phenomena that are explained within the same framework, I'd say.

And as for the second also very interesting question.

If I understood you well, you gave two alternatives.

Either you have pleasure within the all along the whole heart that we have described or you have pleasure in the end.

Or you have pleasure in the end.

Or to explain the relation between pleasure and epistemic arc would it would be to say, well, there's kind of a there's a process theory of active inference.

Right.

So it speaks also to actual neurophysiology, whether that theory holds that's a that's an open question.

You know, that's an open question, but, you know, for the claim to work that computational neurophysiology would have to match the neurophysiology of pleasure.

Right.

I don't know if that's the case or the alternative is, well, it's not about the arc.

It's about the end product.

And the arc is what you need to get to the end product.

Right.

Right.

So, yeah.

Yeah.

It's very interesting.

I agree.

And that's one of the problems that I was hinting at.

And in the end that there is no clear story to my knowledge, but my knowledge is very limited because I have a philosophy, I'm a philosopher by training about how to map these things into what is known about the neurophysiology of pleasure, the limbic system.

There is some some work in that direction.

We mentioned some of it in the in the in the paper that has to do with the dopaminergic discharge during the

harm moment, I guess.

So in the end, that would be in the end of the process.

But conceptually, at least what sustains the attention of the viewer throughout the process is first of all, curiosity is not pleasure or not pleasure in a traditional sense, but the expectation that things are going to make sense in the end if we take enough effort to make to observe them.

And the other half is the pleasure of I would say in which uncertainty, if you are lucky, gets the reduced and gives you this output.

But yeah, I'm curious to just to see what what Sandra thinks about the question as well.

Yeah, I'm not.

I think if you talk about neurophysiology, of course, active inference has a whole other take on the dopaminergic system and all these reward systems.

But but but but but in a sense, like I think the the positive effect is just like you talked about this confidence boost after restructuring after suddenly restructuring.

I think that the the positive effect is just this feeling of the confidence and opening up of of alternatives, which probably has to do with with kind of.

I think that the positive effect is just a precision that is.

Yeah, in a diffuse way.

Protecting back to do all the things that you you are doing, but but in what way specifically that that's that's very unclear to me like that it seems to be that dopamine dopamine might be well placed or these systems might be well placed to have a diffuse effect on further processing in a kind of way.

Yeah, energizing sense.

But but but but but but but but I'm not thinking about pleasure as the traditional reward sense.

I don't I don't think that's very useful in in a in an active inference framing.

Yeah, because yeah, because yeah, it doesn't really explain much if you if you just point to a reward center that is then activated them.

So so so so I I I try to see it as more as as a kind of network effect and as the kind of.

Yes, as you discuss this subsequent effects of of an aha that then boost your your your autonomy by by having these effects on precision.

Yes, but yeah, that's all very speculative still I think.

Yeah.

So so two points Jacopo you said having a sustained attention.

Right.

Right.

So maybe in what we call it aesthetic experience there there needs to be sufficient sufficiently sustained attention for at least for long enough to get to that sort of rewarding but not rewarding point, but that end point center that you're talking about.

Right.

Right.

So maybe that's the them, you know, the the epistemic aspect of that arc.

It's its function would be to maintain the attention for long enough.

And now the question is, you know, what what what does that afford to to that experience is and then an answer could be something like there's research on how in those models, the volatility in your beliefs about the structure of the world.

That is your transition matrix that relate to stuff like anxiety.

Right.

From the point of view of neurophysiology.

Right.

So maybe that what we call aesthetic pleasure is just the release or the the reduction of a feeling of anxiety and uncertainty.

Right.

And the moment is not like a of knowledge.

It's a relief.

Right.

So it's I don't know, maybe maybe that's how you can conciliate.

But at any rate, you would have to make or people who make that claim that the arc is pleasurable because reducing uncertainty is pleasurable.

I again, I'm not that I know center.

That's not that's not your claim.

But I feel like that in the literature, we sometimes read that minimizing the uncertainty per se is pleasurable.

Yeah.

On the time, I doubt that that's the case.

Maybe has no reason that is, you know, reduce anxiety makes you know, you know, in a state that you're poised to, you know, having to sort of pleasurable and partner something like that.

Yeah, that's an important point that maybe wasn't emphasized enough.

But indeed, like if if if active inference is true, then we're constantly reducing uncertainty by actions, by walking.

And so it cannot be merely that it has to do with some kind of unexpected rate of of of uncertainty reduction.

So so they have have to be that's why artists work with with obstacles and and and and creating uncertainty to to to try to have us this experience.

But always in a in a like fallible way, of course, but to try to give us the experience that that we suddenly resolve the uncertainty in a way that is faster than than we expected, maybe when engaging with it first.

Yeah, but that's just that just sort of pushes the problem upward.

Right. It's not a first level of uncertainty in the sort of it's the uncertainty over the uncertainty.

But, you know, if you say, well, that's you know, the pleasure comes from reducing the uncertainty over the uncertainty.

You haven't solved the problem. Right. You just sort of further pushing upward in the hierarchy.

Basically, I guess the problem is that we don't have a good story of why reducing uncertainty per se would be tied to like pleasurable experience.

Right. And what what what what would what would you find like a

kind of satisfying explanation?

Because for me, it's not so much that like.

It seems like this is more puzzling to me to start from a rewards perspective where people get rewarded by some things suddenly by having some activity and reward structures.

The advantage of this story is that that it's not it's not about the reward, but it's about this engagement with the environment that that is positively or negatively balanced depending on how well we are doing in this structuring or predicting it.

So so so. So yeah, I guess maybe your question is then like, what is still the function of the positive experience?

Do we need it at all if it's can it be just an epiphenomenon or something?

And that's a fair question question. But but but but.

Yeah, I think that we have to look at the the consequences of the positive effective experience and and and not so much. Yeah.

And I don't know. Does that does that make sense to you to that kind of or what's your question about the the the phenomenology of of effect or.

Right. I mean, I, I would have I could respond something, but I maybe then there are people that have questions in the chat or do you want to, you know, I think there's like 14 minutes left.

So if you want to. I think this points to some of the the very complex and interwoven aspects of the art viewing experience and separating out like primary sensory reductions of uncertainty, then there's art pieces that leave us with questions.

So then, OK, well, then, OK, well, then that is increasing our precision on us being the kind of cultural thing that's left with a question after the Shakespeare play.

And those start getting into very stacked and and socially enacted kinds of uncertainties and actions and pleasures that may be very distal from the laptop stand.

And it could be like, oh, it's it's giving me a feeling like there was care in the production of the artifact or I know the person who made it.

So then the eye that the parent brings to the child's art piece is different than the art reviewer for an art show.

And so it's almost like an invitation to explore those alternative hypotheses now within a space where we can at least juxtapose them and recombine them because there's so many media specific and and experiential elements that can't necessarily be separated out very cleanly.

Because we're talking about the human reception of art, not a insect's reception of two solid colors or something that that might be more isolatable to elements like ambiguity or luminance or more basic properties of the stimuli.

And then I think the one, the example you showed with a torn canvas, how that's a trace left by a process.

And then the stop sign too.

I mean, in a way, modern art has kind of broken that that fourth wall with a urinal and with a with a blank squares and all these kinds of things that that complexify just the day to day engagements.

And for some people there, it's a bizarre, unnecessary complexification for other people.

It's kind of a re-enchantment with regular objects.

And it just shows how even an object that somebody did not intend to produce as art could still trigger a

deeply aesthetic experience.

And so it's almost like more questions than answers, perhaps.

Nice.

Yeah.

I mean, yes.

One of the advantages of the story we have been presenting, I think, is that active inference offers is this possibility to exploit the hierarchical organization of our predictions.

Of course, as predictions becomes more abstract and are also more difficult to test and to verify.

And we enter into a whole lot of very difficult to pinpoint predictions, which are cultural in nature, that have enough effect of our appreciation of a particular artwork, but we are not quite able to pinpoint.

But the idea is that, for example, with abstract art, a lot of the way in which they function as aesthetic stimulations is by playing with our very higher level expectations about, for example, how an artwork would look like according to a specific art world, according to a specific community of art consumers.

And if Duchamp, for example, could make art by signing a neuronal, it is because he was playing with his very high level expectations about what art is supposed to look like.

It's like a challenge. It could be conceptualized among other things as a challenge to see that specific object as an artwork in the same way in which I suggested that a particular sensory array could be seen as a cow.

So if you manage to do so, to see that specific object as an artwork, you get that insight, which is similar to that sort of low level insight that you got with the cow, but on a higher level.

And yes, so one could exploit this hierarchical story in order to account for different things among them, abstract art as well.

There is a very interesting piece on abstract art in the theme issue by a art historian, Ladislav Kesner.

So that will be another thing to look into, for example.

Yeah.

Yeah.

One comment from the live chat, Holly wrote,

When thinking about abstract painting, the viewer might find pleasure in compositional elements like variety, contrast, repetition, unity and balance in addition to the interpretation and gesture.

Yeah, sure.

I mean, of course, there are also these things about the arrangement of elements in the canvas, the way in which they create a balanced or unbalanced design, how we explain the intentions of the painter to position these elements in this particular way.

Of course, if we know more about this particular poetic of that artist, if you, for example, it becomes vital with certain artists like Mondrian or Kandinsky to know also about their theoretical writings in order to make sense.

Or why they are using certain colors and certain shapes in certain forms.

So it's also a case that with abstract art, what is sometimes called the broader context, the epitext, all the things, the information around the artwork, gain more importance.

Because they are ways in which artists set expectations on us, not by things that they put in the artwork themselves, but by presenting them within a certain context of theories and so on.

So a lot of predictions are installed by artists in abstract art in this epitext more than in the work itself.

Awesome.

If I can quickly add on because this is a point I wanted to raise is that now we're talking about aesthetic experience.

I think you can appreciate art outside of aesthetic experience, as you said, Jacopo.

So you can appreciate, let's say, an artwork as a art historical object and then you have this sort of intellectual pleasure with it.

And even when you look at, let's say, conceptual art, right?

There's this, well, what will happen oftentimes is that they will use this device called a citation, right?

Whereas in the piece of the artwork, there will be a citation to an earlier work and you can appreciate the citation if you know the art history.

Right. So, but in that case, you're kind of outside. I would, well, and that's the beauty of active inference. Are you really outside of the domain of aesthetic experience?

Maybe you can make a claim that you're within the same sort of regime of like computational processes and concrete processes.

Sure.

Sure. But let's say in the common sense of aesthetic experience, then your experience relies much less on the sort of rolling, sensory processing dynamics, right?

That you would have in something like, I don't know, you were mentioning abstract and, you know, Kenden's key.

And we have a lot of colors. You have difficulties interpreting the figure and so on and so forth. Right.

Maybe it would be useful to talk of different aesthetic experience for different objects and different ways of appreciating those objects.

And that might help clarifying, right?

That may be different, you know, trajectories on the epistemic arc, right? And so on and so forth.

So all the kind of problems that we discussed today, maybe that would be a good starting point with a taxonomy like that.

Cool. Yeah. I mean, I don't know whether Sander agrees, but I think we are always in the same sort of aesthetic,

domain. There are no differences, if not in this sort of abstractions in the nature of predictions that you put into play in the different cases between appreciating something for its low level features and the way in which they are combinable.

And having this very abstract high level knowledge of the art scene that informs your guesses and your appreciation of artworks.

Also because these processes and these guesses inform one another.

So you need to have some plastic view of where the boundary is or not having boundaries at all between these two things.

Because for example, if you know, for example, the title of Picasso bull's head, the thing is of sculpture formed by pieces of bicycle.

Or a bicycle. If you know the side of this sort of a high level information about the artwork.

We are also inclined to concentrate on certain sensory features of the object instead of others and to make

some low level inferences instead of others.

And if you make those inferences, perhaps you formulate new high level hypotheses and these two things continuously inform one another.

And that's the same process to me, as long as you have these epistemic arcs that leads you towards some sort of discovery, which could be very high level or low level.

I mean, to me, it's one of the advantages of the framework. I don't know whether Sander agrees or not.

Yeah, completely. Yeah, I would even make it more broadly.

I mean, even in games, you have these epistemic arcs and just in everyday life, they might.

I mean, we call them like a minimal unit of sense making and it is it is kind of that.

So it is not really necessarily attached to art, but it is a kind of pleasure that is not derived from like food or sex or the kind of typical rewards that people use in psychology.

Psychology. So so so so and that's the advantage, of course, of active inference that it that it makes room for this kind of like I would call it a pleasure of obstacle clearing in a general sense to towards what you want instead of getting what you want.

It's like we want to problem solve in order to get where we want.

And I think it makes sense for evolution to reward like this, this obstacle clearing and this this overcoming challenges, which is, of course, uncertainty.

So overcoming uncertainty is is the reward as such.

And and and it makes sense, I think, for evolution to to to to organize us as as such and maybe as humans more than than other animals.

But you see them also like I give the example of my dog that that often picks the largest kind of unwieldy stick to to play with.

And I see that happen and I I I I think it's probably because it gives him the most challenge and it gives him an obstacle to overcome.

And if I don't want to play with him, he has to find his own obstacles and he has to find ways to reduce uncertainty in this play that that that that he generates himself.

And that's art as well, of course.

So, so, so, so, yeah, that's for me, the, the, the strength of the approach enacted inference.

Yeah, if I can add a very brief remark that that's in to some extent that's going back to the sort of modern origin of aesthetics to Baumgart and who used to think of aesthetics as the science of sense, sense, sense, so, sense, so cognition is that in general.

So, of course, the difference between everyday experience and an aesthetic experience will be that aesthetic experiences or aesthetic objects present ourselves with these more idealized form of uncertainty minimization that we tended to characterize with the epistemic arc.

And I showed how things could go wrong in many ways.

Ideally not when it went on when art works, we get these epistemic arcs in a very purified and idealized form.

And that's what why these experiences are more pleasurable.

Pleasurable, but indeed the, the component, the, the inferential component of that experience is something that we find in experience more broadly because as, as our experience is finding structure in our sensorium, but not all experience is as successful in, in finding structure in our sensorium as aesthetic experiences are.

Is that right?

Yeah.

Which is what he does is super interesting.

And for the in a couple of second, the short story is that's one artist that I discussed in my mastery thesis that I did under supervision of Sander, actually.

So it's interesting for me to this whole art stream because it goes full circle for me.

It started with the aesthetic experience of all like experience and transformative experience and pretty good processing.

Right.

Then move to evolution biology and psychiatry and so on and so forth.

But yeah.

So having invited artists, that would be cool.

And, you know, maybe pursuing those other topics that in the end all relate to aesthetic experience, you know, in biology and so on and so forth.

Yeah.

Awesome.

I hope we can do it.

If someone's listened this far and they want to jump on for an art stream or suggest something, that would be great.

So thank you, fellows.

Great to hear this.

And until next time.

Bye.

Bye.

Bye.

Bye.

Bye.